App Note – HVLP Alpha-Beta Update

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1 Introduction

1.1 Purpose and Scope

This document provides guidance for hardware\software changes when migrating from HVLP Alpha to HVLP Beta inverter builds.

1.2 Document Conventions and Terminology

Typeface	Indicates
WARNING	Very important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent serious problems.
CAUTION	Important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent undesirable problems.
NOTE	Useful information for the document user.

Table 1. Typographical Conventions

1.3 Glossary

Term	Description
DCF	Device Configuration File: CANopen settings file
EDS	Electronic Data Sheet: CANOpen interface file used to interpret object descriptions, values and scalings.

Table 2. Glossary

2 Background Information

2.1 Reason for Update

As part of the HVLP development lifecycle several hardware and software modifications have been implemented.

2.2 Overview of Changes

The following changes have been made to the HVLP and are summarised below

1. The DC Link capacitors have been changed from electrolytic type to film type. The means the total DC link capacitance has now changed from 450µF to 120µF
2. The logic supply storage capacitor was increased in value, this is to allow the unit to operate correctly during a 10ms power outage of the KL30 supply
3. The I/O connections have been changed to provide consistency with Borg Warner Gen 5 motor controllers
4. The HVIL circuitry has been simplified. The excitation circuit was removed and the input circuit has been changed.
5. The UVW and AB encoder inputs have been changed from differential to single ended type
6. The sin/cos encoder inputs have been combined with the AB encoder inputs and use the same pinouts.
7. An additional motor thermistor input was added.
8. The off state current (with KL15 low) has been reduced to below 1mA

2.2.1 Hardware Changes

Part of the hardware updates has involved a new PCB design which has resulted in several pinout allocations changing from the Alpha hardware builds. The following table details the pinouts changes between hardware builds.

Pin Out	HVLP Alpha Pin Outs	HVLP Beta Pin Outs
A4	CAN 1 L a	CAN 1 H a
A3	~Enc B	Digital Input 1
A2	Enc B	CAN 1 H b
A1	Resolver Cos Lo	Encoder Vcc
B4	CAN 1 L b	Encoder 0V
B3	~Enc A	Resolver Excitation -
B2	Enc A	Digital Input 2
B1	Resolver Cos	Resolver Excitation +
C4	CAN Termination	CAN 1 L a
C3	~Enc W	CAN Gnd
C2	Enc W	CAN 1 L b
C1	Resolver Sin Lo	CAN Termination
D4	CAN 1 H a	Not Connected
D3	~Enc V	Diigital Input 4
D2	Enc V	Digital Input 3
D1	Resolver Sin	Ain 1 Vcc
E4	CAN 1 H b	Not Connected

E3	~Enc U	Resolver Cos
E2	Enc U	Enc U
E1	Resolver Excitation -	Ain 1
F4	CAN Gnd	Ain 2
F3	Encoder 0V	Resolver Cos Lo
F2	Encoder Vcc	Not Connected
F1	Resolver Excitation +	Enc A\ Sin
G4	HVIL	HVIL A
G3	Cos	Thermistor 2
G2	Sin	Ain Gnd
G1	HVIL	Thermistor 1
H4	Diigital Input 4	HVIL B
H3	Digital Input 2	Resolver Sin
H2	Ain 3 Thermistor	Not Connected
H1	Ain 1 Vcc	Enc V
J4	Digital Input 3	AIN 3
J3	Digital Input 1	Resolver Sin Lo
J2	Ain 2b	Enc B\Cos
J1	Ain 2a	Ignition Kl15
K4	Mot Connected	Dout 4 Source
K3	Not Connected	Dout 3 Source
K2	Ain 1b	Ain 4
K1	Ain 1a	Enc W
L4	Logic 0V	Logic 0V KL31
L3	Dout 3	Not Connected
L2	Logic 0V	Not Connected
L1	12-24 Supply KL30	12-24V Supply KL30
M4	Logic 0V	Logic 0V KL31
M3	Dout 4	Not Connected
M2	Ignition Switch KL15	Not Connected
M1	12-24 Supply KL30	12-24V Supply KL30



WARNING: It is important to verify that the pinouts of the system are compatible with the hardware build of the HVLP. Failure to verify this could result in product damage.

In Alpha hardware builds if using AB or UVW encoders the setup of AB & UVW complimentary encoders was also mandatory. This is no longer required in Beta hardware builds meaning any existing encoders must be changed accordingly and any external hardware to provide the complimentary encoder inputs is no longer required.

2.2.2 Firmware Changes

As part of the HVLP Beta release it has also been necessary to create a separate Beta hardware software version. At the time of writing this Application Note the following are the existing Alpha\Beta firmware versions

	Alpha	Beta
IOP SW Version	SN0089.51	2206.0001 (Released Version Number)
MCP SW Version	SN0088.54	2104.0001 (Released Version Number)

All HVLP products are delivered with correct hardware compatible IOP & MCP firmware which means it should not be necessary to update your HVLP firmware unless the application requirements need specific updates not available in this version, or a newer version is available. If you have any doubt please contact your local BorgWarner representative.

2.2.3 Configuration Changes

On first setup of a migrating Alpha-Beta system it is more than likely that a new configuration setup will be required. If you have an existing DCF (Device Configuration File) settings file for your Alpha setup then this can be downloaded into your Beta product and used a starting point for the system setup. Additional parameters may be required to be setup after this however, if you have any difficulties then please contact your local BorgWarner representative.

2.3 DVT Update

DVT has been updated to fully support the latest Beta firmware, to make use of this please ensure your DVT and EDS file is up to date.

2.4 Product Manual

A specific product manual has been created for HVLP Beta builds. Contact your local BorgWarner representative for a copy.